

Project Title

Mini Project: Vulnerability Scanner

Objective

To develop a simple Python-based vulnerability scanner that detects common open ports, identifies basic security weaknesses, and generates a vulnerability report.

Tools Used

- Python 3
- Socket Library
- Python IDLE / VS Code

Features

- Scans common network ports
- Detects open services
- Suggests possible vulnerabilities
- Generates a text report automatically

Working

1. User enters an IP address.
2. The program scans common ports (21, 22, 25, 80, 443, 3306).
3. If a port is open, it displays the service name.
4. The program checks for common security issues related to that service.
5. A report is saved as report.txt.

Source Code

```
import socket
```

```
ports = {
```

```
    21: "FTP",
```

```
    22: "SSH",
```

```

25: "SMTP",
80: "HTTP",
443: "HTTPS",
3306: "MySQL"
}
target = input("Enter IP Address: ")
print("\nScanning...\n")
report = []
for port, service in ports.items():
    s = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
    s.settimeout(1)
    result = s.connect_ex((target, port))
    if result == 0:
        print(f"[OPEN] Port {port} ({service})")
        report.append(f"Port {port} ({service}) : OPEN")
    s.close()
with open("report.txt", "w") as file:
    for line in report:
        file.write(line + "\n")
print("Report saved as report.txt")

```

Expected Outcome

The scanner helps identify open network ports and highlights possible vulnerabilities, giving beginners an introduction to vulnerability assessment and penetration testing concepts.